

# Saurav Kumar

+916206846984 | sauravx25@gmail.com | linkedin | github | portfolio

## Education

National Institute of Technology, Warangal

Nov 2021 - May 2025

Bachelor of Technology in Electronics and Communication Engineering

CGPA: 7.69/10

- Relevant Coursework: Data Structures and Algorithms, Operating Systems, OOP, DBMS

## Experience

SAP Labs India

July 2025 - Present

Associate Developer

Bengaluru, India

- Built **CAP backend with OData services** and **Spring Boot APIs**, orchestrating **600+ automated tests** across **5 SAP HANA clusters** with dynamic dependency handling
- Implemented locking and DB isolation for **200+ concurrent runs**, reducing failures by **30%** and cycle time by **50%**
- Architected a **multi-target, API-first cloud framework** on **SAP BTP** registered in the **SaaS Registry**, enabling scalable execution across environments
- Added async **GenAI-based failure classification** and auto-healing (**85% accuracy**), reducing manual triage by **60%**
- Developed a **React + UI5 Web Components UI** with Dashboards and Custom Reports for test analysis, centralized configuration system, and branch-based execution

SAP Labs India

May 2024 - July 2024

SDE Intern

Bengaluru, India

- Built a deepfake detection pipeline (EfficientNet-B4) trained on **140k+ samples**, achieving **98% validation accuracy**
- Built a Flask inference service supporting **300+ concurrent requests** with real-time image/video processing; optimized 1080p inference to **22 FPS**
- Tech: **Python, TensorFlow, Flask, OpenCV, WebRTC, React**

## Projects

Railway Ticket Booking System

[GitHub](#)

- Designed a **Spring Boot application** with PostgreSQL, Redis, Apache Kafka, and Elasticsearch to handle high-concurrency train bookings
- Built a **distributed seat allocation system** using Redis Lua scripts and TTL-based locks, combined with idempotent APIs and optimistic concurrency control to prevent race conditions and overbooking
- Designed an **event-driven workflow** using Kafka (choreography) for cancellations, refunds, and waitlist promotion with retry handling and dead-letter queues
- Developed a **CQRS-based search pipeline** (PostgreSQL to Elasticsearch sync) with Redis caching, achieving sub-second train search latency
- Secured services using **JWT-based RBAC**, exposed 30+ REST APIs via Swagger, and built a frontend using React and shadcn/ui for booking and search flows

## Technical Skills

Languages: **Java, C++, JavaScript, Python, SQL**

Backend & Frameworks: **Spring Boot, Node.js, SAP CAP, Express, Elasticsearch**

Databases & Messaging: **PostgreSQL, MySQL, Redis, Kafka**

Cloud & DevOps: **Docker, Kubernetes, AWS (EC2, S3, SQS), Jenkins, Git**

Core Concepts: **Microservices, Concurrency, OOP, DSA, Networking**

## Achievements

- 99.32%ile** among **10+ lakh** candidates in **JEE Mains (Mar 2021)**.
- NTSE Qualified** and **District Topper** in Class X(97.6%) also secured 95.8% in Class XII (CBSE)
- Solved **500+ algorithmic** problems on LeetCode, Codeforces, and InterviewBit.